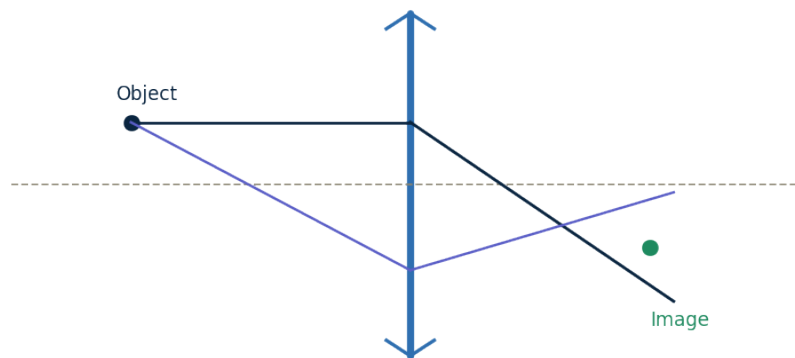


Optics

JEE Physics — Notes, Formula Sheet, Practice Problems & Diagram

1. Notes

- Reflection: angle of incidence = angle of reflection. Refraction follows Snell's law: $n_1 \sin(\theta_1) = n_2 \sin(\theta_2)$.
- Lens/mirror formula: $1/v - 1/u = 1/f$ (lens, sign convention applied) or $1/v + 1/u = 1/f$ (mirror, with sign convention). Magnification $m = v/u = h(\text{image})/h(\text{object})$.
- Power of a lens $P = 1/f$ (in meters), measured in diopters (D). For lenses in contact, $P(\text{total}) = P_1 + P_2 + \dots$
- Total internal reflection occurs when light travels from a denser to a rarer medium at an angle greater than the critical angle: $\sin(\theta_c) = 1/n$.
- For a prism, the angle of deviation depends on the refractive index and prism angle; minimum deviation occurs at a specific angle of incidence, giving $n = \sin((A+D_m)/2) / \sin(A/2)$.
- Wave optics: Young's double-slit experiment shows interference; fringe width $\beta = (\lambda D)/d$, where D is the screen distance and d is the slit separation.
- Diffraction is the bending of light around obstacles/slits; single-slit diffraction minima occur at a $\sin(\theta) = n \lambda / a$.
- The human eye's power of accommodation lets it focus on objects from the near point (about 25 cm) to the far point (infinity, for a normal eye).



Ray diagram through a convex lens: rays from the object converge to form a real image.

2. Formula Sheet

Snell's law	$n_1 \sin(\theta_1) = n_2 \sin(\theta_2)$
Lens formula	$1/v - 1/u = 1/f$
Mirror formula	$1/v + 1/u = 1/f$
Magnification	$m = v/u = h_i/h_o$
Power of a lens	$P = 1/f$ (in meters, unit: diopter)
Critical angle	$\sin(\theta_c) = 1/n$
Young's double slit fringe width	$\beta = \lambda D / d$
Single slit diffraction minima	$a \sin(\theta) = n \lambda$

3. Practice Problems (22 questions)

1. A convex lens has a focal length of 20 cm. An object is placed 30 cm from the lens. Find the image distance and magnification.
2. Find the power of a lens with a focal length of 0.5 m.
3. Light travels from air ($n=1$) into glass ($n=1.5$) at an angle of incidence of 30 degrees. Find the angle of refraction.
4. Find the critical angle for total internal reflection from glass ($n = 1.5$) to air.
5. In Young's double slit experiment, the slit separation is 0.5 mm, the screen distance is 1 m, and the wavelength is 500 nm. Find the fringe width.
6. A concave mirror has a focal length of 15 cm. An object is placed 10 cm in front of it. Find the image distance.
7. Two thin lenses of focal lengths 20 cm and 25 cm are placed in contact. Find the focal length and power of the combination.
8. State the two laws of refraction (Snell's law) and explain what determines the refractive index of a medium.
9. A ray of light passes through a prism of angle 60 degrees and undergoes minimum deviation of 30 degrees. Find the refractive index of the prism material.
10. An object is placed at the focus of a convex lens. State where the image is formed and describe its nature.
11. Explain why a diamond sparkles more than an ordinary glass gem, in terms of its critical angle.
12. A single slit of width 0.2 mm is illuminated with light of wavelength 600 nm. Find the angular position of the first diffraction minimum.
13. A concave lens has a focal length of -25 cm. An object is placed 50 cm from the lens. Find the image distance.
14. State the difference between a real image and a virtual image, with one example of each.
15. The near point of a person's eye is 50 cm (instead of the normal 25 cm). Find the power of the corrective lens needed to enable near vision at 25 cm.

16. A ray of monochromatic light passes from a denser to a rarer medium and undergoes total internal reflection. State the condition necessary for this to happen.
17. In a Young's double slit experiment, if the wavelength of light used is doubled, how does the fringe width change (other parameters constant)?
18. Explain, with a ray diagram description, how a simple magnifying glass produces a magnified virtual image.
19. Two convex lenses of focal length 10 cm each are placed 5 cm apart along the same axis. An object is far away (at infinity). Describe qualitatively where the final image forms relative to the second lens.
20. State Huygens' principle and briefly explain how it is used to derive the laws of reflection and refraction.
21. A microscope objective has focal length 1 cm and the eyepiece has focal length 5 cm; they are separated by 20 cm. Explain briefly what determines the overall magnification of a compound microscope.
22. Explain why the sky appears blue during the day, based on the scattering of sunlight.

Answer Key

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| 1. $v = 60$ cm, $m = -2$ (image inverted, magnified) | 2. $P = 2$ D |
| 3. θ_2 approx 19.5 degrees | 4. θ_c approx 41.8 degrees |
| 5. $\beta = 1$ mm | 6. $v = -30$ cm (virtual image behind the mirror) |
| 7. $f(\text{combined})$ approx 11.1 cm, $P = 9$ D | 8. Conceptual - Snell's law: $n_1 \sin(\theta_1) = n_2 \sin(\theta_2)$; n depends on the medium's optical density and light's wavelength |
| 9. $n = 1.5$ (approx, using $n = \sin((A+D_m)/2)/\sin(A/2)$) | 10. Conceptual - image forms at infinity; rays emerge parallel (used in projectors, telescopes) |
| 11. Conceptual - diamond has a very high refractive index, giving a small critical angle, so more light undergoes total internal reflection, causing more sparkle | 12. θ approx 0.172 degrees (small angle, $\sin(\theta) \approx \lambda/a$) |
| 13. v approx -16.7 cm (virtual image, same side as object) | 14. Conceptual - real image can be formed on a screen (e.g., cinema screen); virtual image cannot (e.g., image in a plane mirror) |
| 15. $P = +2$ D | 16. Conceptual - angle of incidence in the denser medium must exceed the critical angle |
| 17. Fringe width doubles (β proportional to λ) | 18. Conceptual - object placed within focal length of convex lens; lens forms an enlarged, virtual, upright image viewed by the eye |
| 19. Conceptual - final image forms further forward, closer to (or possibly beyond) the second lens depending on exact geometry; acts as a compound lens system | 20. Conceptual - each point on a wavefront is a source of secondary wavelets; used to derive both laws geometrically |
| 21. Conceptual - overall magnification = magnification by objective $\times$ magnification by eyepiece | 22. Conceptual - shorter (blue) wavelengths scatter more strongly in the atmosphere (Rayleigh scattering), so blue light reaches our eyes from all directions in the sky |

Note: conceptual/derivation questions are marked accordingly in the answer key. Numerical answers are rounded for clarity — show full working in your own solutions.